

Boishik Barua

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Summary

Computer Science Graduate with hands-on experience in machine learning, computer vision, NLP, full-stack development, and software engineering. Skilled in Python, PyTorch, TensorFlow, Scikit-learn, JavaScript, Node.js, Express.js, and MySQL. Built AI, research, and web-based projects involving text-to-face generation, hallucination detection, medical image classification, recommendation systems, and multi-role e-commerce platforms.

Education

North South University <i>B.Sc. in Computer Science and Engineering</i>	Dhaka, Bangladesh <i>Expected Jun 2026</i>
• CGPA: 3.59/4.00 (89.5%); Distinction: Cum Laude, 25% Scholarship Holder	
Notre Dame College <i>Higher Secondary Certificate, Science</i>	Dhaka, Bangladesh <i>2020</i>
• GPA: 5.00/5.00	
Monipur High School <i>Secondary School Certificate, Science</i>	Dhaka, Bangladesh <i>2018</i>
• GPA: 5.00/5.00	

Technical Skills

Languages	Python, C, C++, Java, JavaScript, SQL
ML / AI	Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, CNNs, Vision Transformers, NLP, Knowledge Distillation, Grad-CAM
Web / Backend Tools	HTML, CSS, JavaScript, Node.js, Express.js, MySQL
Concepts	Git, GitHub, Jupyter Notebook, Google Colab, XAMPP, AutoCAD
Office Tools	Computer Vision, Natural Language Processing, Feature Engineering, Model Evaluation, OOP, Data Structures and Algorithms
	MS Word, MS Excel, PowerPoint, Google Docs, Google Sheets, Google Drive, PDF tools

Awards & Certifications

1st Runner-Up – Innovation Challenge Season 19 <i>North South University</i>	2026 <i>Dhaka, Bangladesh</i>
• Awarded 1st Runner-Up and Best Poster Design for “Face Reconstruction using Hierarchical Geometry Conditioning from Witness Description.”	
GCL World 2026 – AI and Data Science Training Program <i>Matsuo Iwasawa Lab, University of Tokyo</i>	Apr 2026 – Present <i>Japan</i>
• Completed intensive training on practical and theoretical implementation of AI, machine learning algorithms, and data science.	
Intra NSU Junior Programming Contest – Summer 2023 <i>North South University</i>	Summer 2023 <i>Dhaka, Bangladesh</i>
• Ranked 8th and achieved the second-highest number of solved problems.	

Research Projects

Forensic Text-to-Face Suspect Generation (Journal Submitted) 2025 – Present
Capstone Project

- Developing a forensic AI system that generates suspect-like facial images from witness text descriptions.
- Applying Stable Diffusion, ControlNet-based conditioning, and geometry-guided generation to improve facial structure consistency.
- Combining natural language descriptions with hierarchical facial geometry cues for controlled image synthesis.

Bangla Hallucination Benchmark for Summarization 2025 – Present
Research Project

- Contributing to a benchmark for hallucination detection in Bangla text summarization.
- Experimenting with BanglaBERT, XLM-R, multilingual encoders, and LLM-based baselines for hallucination classification.
- Supporting dataset preparation, model evaluation, and comparative analysis for low-resource NLP.

Hybrid Feature-Fusion Network with Knowledge Distillation (Conference Submitted) 2025
Pattern Recognition Project

- Built a dental X-ray classification pipeline using ConvNeXt, Swin Transformer, ResNet34, and knowledge distillation.
- Integrated Grad-CAM to visualize model attention and improve interpretability of predictions.
- Designed a hybrid feature-fusion approach for medical image classification.

Academic Projects

Baggy – Multi-Role E-Commerce Platform 2025
Full-Stack Web Project

- Developed a full-stack e-commerce platform using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL.
- Implemented product browsing, cart, wishlist, stock management, order tracking, voucher logic, and payment-status workflows.
- Built database-backed customer, seller, and support role features for multi-user platform functionality.

BanglarKrishok – Agricultural Expert System Website 2025
Software Engineering Project

- Built a rule-based agricultural recommendation system using Node.js, Express.js, MySQL, JavaScript, and JSON-based crop rules.
- Served as second team lead and contributed to system design, UI mockups, and UML documentation.
- Prepared use case, class, sequence, activity, and state diagrams for software design analysis.

Movie Recommender System 2025
Machine Learning Project

- Developed and deployed an interactive movie recommendation system using a curated dataset.
- Implemented search-based recommendations using movie title and genre keywords.
- Added a watchlist feature to help users save movies for later viewing.

Teaching Experience

Private Tutor 2021 – Present

- Delivered academic instruction for SSC and HSC students over 5 years.
- Subjects taught: SSC (Physics, Chemistry, Mathematics), HSC (Physics, Mathematics).